Machine Learning Based Clinical Trajectory Prediction in Emergency Care: A Performance Study Using the MC-MED Dataset

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ABSTRACT

Background: Emergency department crowding is a critical public health challenge. Existing machine learning prediction models are limited by fragmented outcome focus, late-feature dependencies, and insufficient interpretability validation. Critically, few existing studies have established a framework to distinguish whether features identified as "important" via Shapley Additive exPlanations (SHAP) are causally necessary for prediction, merely redundant, or paradoxically harmful.

Methods: We developed and evaluated 12 machine learning algorithms spanning five architectural families to predict interrelated clinical trajectories using only admission-available variables from the MC-MED dataset (118,385 adult emergency department visits, 2020-2022): (1) emergency department length of stay (short ≤ 4h, medium 4-12h, prolonged > 12h), (2) hospital admission (binary), and (3) hospital length of stay among admitted patients (short ≤ 3d, medium 3-7d, prolonged > 7d). The primary analytic contribution was a feature classification framework distinguishing predictive features (removal causes ≥ 5 percentage point performance degradation), redundant features (removal causes < 2 percentage point change), and harmful features (removal paradoxically improves performance by ≥ 3 percentage points). Bootstrap confidence intervals (1,000 resamples) were calculated for all performance metrics.

Results: A feature ablation study revealed three distinct patterns of predictor behavior across 12 machine learning architectures. First, predictive features: removing triage acuity reduced admission prediction sensitivity by 7.1 to 8.4 percentage points across all models. Second, redundant features: removing emergency medical services arrival produced minimal sensitivity changes for tree-based models (± 0.4 percentage points) but moderate decreases for Logistic Regression (-1.3 percentage points), indicating model-dependent importance where tree-based methods captured correlated information. Third, harmful features: removing Charlson Comorbidity Index paradoxically improved hospital length of stay sensitivity by 16.0 percentage points in LightGBM and 4.4 percentage points in Logistic Regression, with no effect on Random Forest. Architecture-agnostic Shapley Additive exPlanations validation demonstrated 75-79 percent Jaccard similarity for top-5 features across model families. For multiclass prediction tasks (emergency department length of stay and hospital length of stay), we prioritized sensitivity for the high-risk class because missing a patient at risk of prolonged stay has greater clinical consequences than false alarms. For binary admission prediction, we prioritized area under the receiver operating characteristic curve (AUROC) because it balances sensitivity and specificity across all thresholds, appropriate for resource allocation decisions. Using this task-specific metric framework, LightGBM achieved the highest sensitivity for prolonged emergency department stay (82.0 percent, 95 percent confidence interval: 80.7 percent to 85.7 percent). XGBoost achieved the highest area under the receiver operating characteristic curve for admission prediction (0.809, 95 percent confidence interval: 0.802 to 0.817). Random Forest achieved the highest sensitivity for prolonged hospital stay (73.5 percent, 95 percent confidence interval: 70.9 percent to 76.1 percent). Narrow confidence interval widths (2.8 percent to 5.2 percent) confirmed stable performance across all models. Demographic-specific threshold adjustment reduced racial sensitivity disparities from 9.1 to 0.5 percentage points.

Conclusion: A feature classification framework distinguishing predictive, redundant, and harmful features provides evidence-based guidance for feature selection in emergency department machine learning. Integrative predictive modeling of interrelated emergency department trajectories using only admission-available variables is feasible and robust. Bootstrap confidence intervals confirm stable model performance that would generalize to similar populations. Tree-based gradient boosting models offer an optimal balance of predictive performance and computational efficiency for real-time deployment.

Keywords: machine learning, emergency department, clinical trajectory, length of stay, feature selection, feature ablation, Shapley Additive exPlanations stability, health equity, triage, interpretability, confidence intervals

1 INTRODUCTION

Emergency department (ED) crowding represents one of the most pressing public health challenges worldwide, with profound implications for patient safety, healthcare costs, and quality of life [1, 2]. Prolonged ED length of stay (ED-LOS), particularly boarding times exceeding 12 hours, has been associated with a 37% elevated mortality risk among admitted patients [5, 58]. These challenges underscore the critical need for accurate, real-time prediction tools to inform triage decisions and optimize resource allocation.

To understand ED crowding conceptually, Figure 1 presents the Input-Throughput-Output model proposed by Asplin et al. [3]. This framework identifies three interdependent stages where bottlenecks occur: *input* factors (patient volume, acuity, emergency medical services arrivals, walk-ins), *throughput* factors (triage efficiency, diagnostic testing, consultation availability), and *output* factors (inpatient bed availability, discharge planning, observation/transfer). The model highlights that crowding is not a single-point failure but a systemic problem requiring interventions across all three stages. Our study focuses on the throughput stage-specifically, using machine learning (ML) to accelerate triage and disposition decisions.

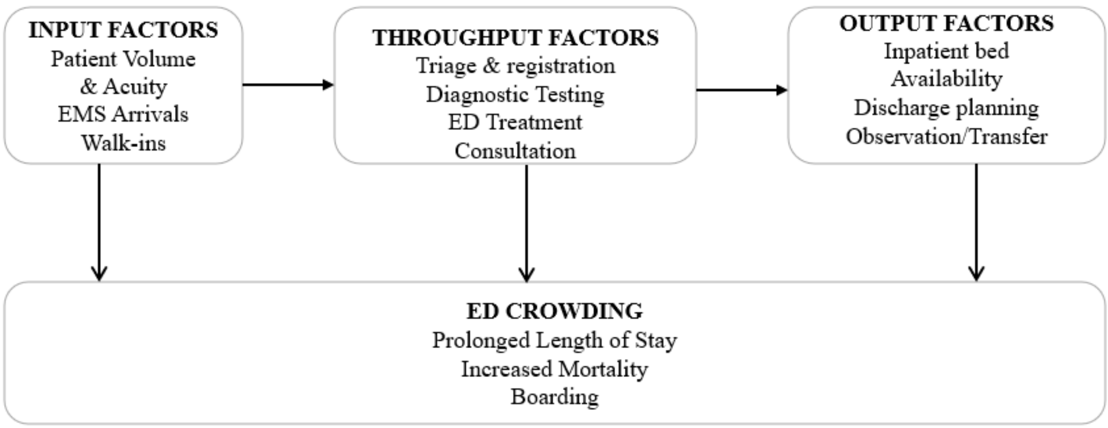

Figure 1. The Input-Throughput-Output Model of Emergency Department Crowding. This framework illustrates how factors at three stages-input, throughput, and output-interact to create emergency department crowding. Interventions targeting any single stage are insufficient; systemic solutions are required. Adapted from Asplin et al. [3].

Machine learning has emerged as a promising approach for clinical trajectory prediction in emergency care. Recent studies demonstrate ML's potential to predict ED-LOS, hospital admission, and hospital length of stay (Hosp-LOS) [7, 8, 16]. However, the existing literature has four critical limitations.

First, most studies examine outcomes in isolation rather than as interconnected components of a clinical continuum [7]. The majority of existing work predicts emergency department length of stay alone, with fewer than half examining hospital admission, and even fewer predicting hospital length of stay among admitted patients. This fragmented outcome focus fails to reflect clinical reality. As shown in Figure 2, conventional studies (Panel A) predict a single outcome (typically emergency department length of stay) while ignoring subsequent outcomes such as admission or hospital length of stay. This ignores that triage acuity influences admission likelihood, which in turn determines inpatient resource utilization. In contrast, our integrative outcome framework predicts the full clinical cascade of three interrelated outcomes: emergency department length of stay, hospital admission, and hospital length of stay. Although we use separate, task-optimized models for each outcome, our contribution lies in the integrated prediction of the complete clinical trajectory rather than isolated outcomes.

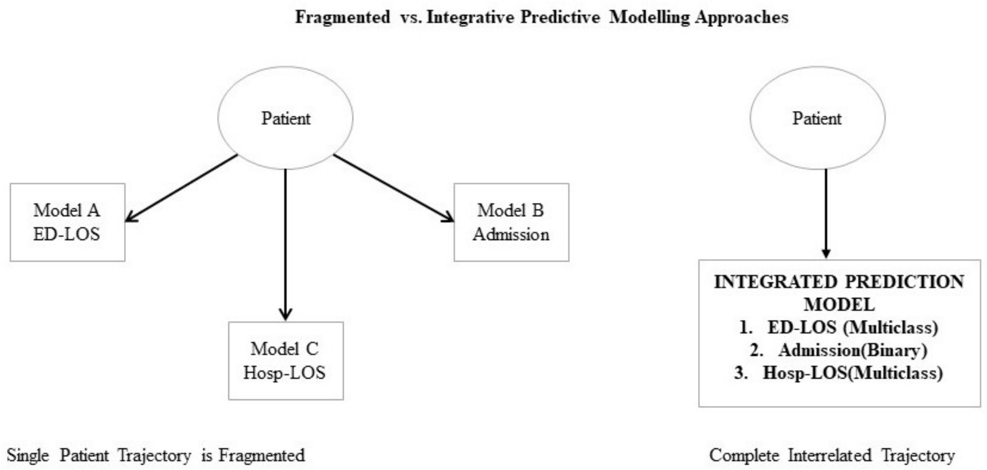

Figure 2. Comparison of prediction strategies. (A) The fragmented outcome approach prevalent in existing literature, where studies predict a single outcome (typically emergency department length of stay) and ignore subsequent outcomes such as hospital admission or hospital length of stay. (B) The integrative outcome framework of this study, which predicts the full clinical cascade of the interrelated outcomes using task-optimized models for each outcome.

Second, many existing models incorporate late-feature dependency, utilizing laboratory results, specialist consultations, or disposition plans unavailable at triage, limiting real-time applicability [11].

Third, insufficient interpretability validation, typically relying on single-model Shapley Additive exPlanations (SHAP) analysis, raises concerns about algorithm-specific artifacts versus true clinical signals [12, 13].

Fourth, few existing studies have systematically evaluated whether features identified as ”important” via Shapley Additive exPlanations are causally necessary for prediction, merely redundant, or paradoxically harmful. Conventional feature importance ranking often assumes that higher importance equates to greater predictive value, but this assumption has received limited empirical testing. Consequently, clinicians and model developers have limited evidence-based guidance for feature selection, retention, or removal in emergency department prediction systems. A framework to distinguish predictive, redundant, and harmful features would help address this gap.

This study addresses these gaps through a feature-centric evaluation of 12 machine learning algorithms on the MC-MED dataset (118,385 adult emergency department visits, Stanford Health Care, 2020-2022) [5]. The primary contribution is a systematic framework for classifying features into predictive, redundant, and harmful categories, offering evidence-based guidance for feature selection in emergency department prediction systems. Three secondary objectives provide methodological rigor: (1) architecture-agnostic interpretability validation using Jaccard similarity across model families [12, 46], (2) demographic disparity analysis with threshold adjustment [37, 13], and (3) severity-stratified clinical utility assessment [3, 4].

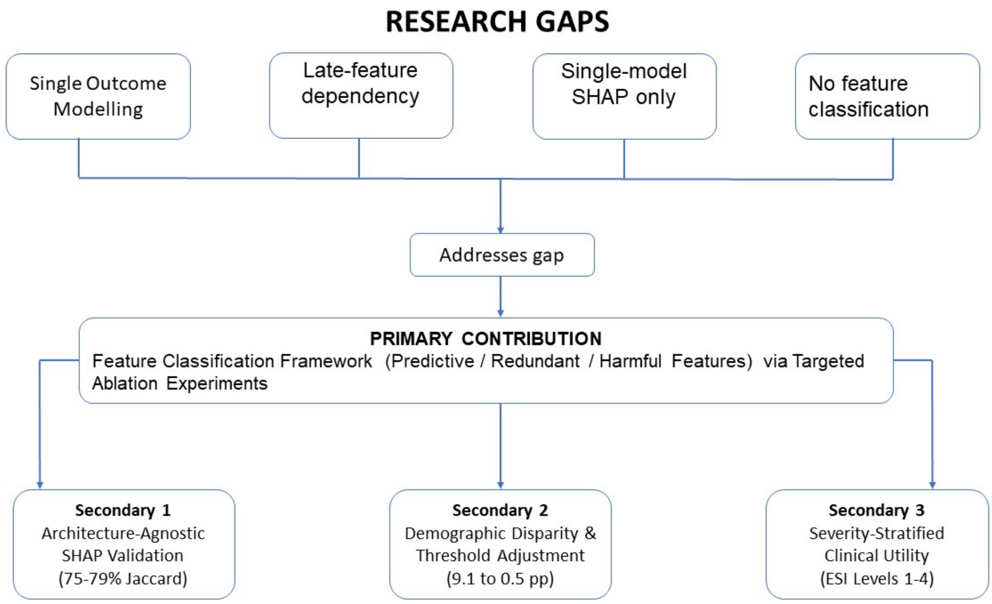

Figure 3. Research framework of this study. The primary contribution is a feature classification framework that distinguishes predictive, redundant, and harmful features via targeted ablation experiments. This framework is then applied to three secondary objectives: (1) architecture-agnostic SHAP stability validation across model families, (2) demographic disparity analysis with threshold adjustment, and (3) severity-stratified clinical utility assessment by Emergency Severity Index level (ESI Levels 1-4; Level 5 non-urgent visits were excluded due to very low admission rates [¡0.5% of admissions]).

Finally, Figure 4 presents the complete conceptual framework guiding our analysis. The framework flows from the MC-MED dataset through admission-available variables and twelve machine learning algorithms to three interrelated prediction tasks, culminating in a comprehensive evaluation framework. This structure positions the interrelated clinical trajectories within an integrative emergency care framework, from triage through disposition to inpatient care.

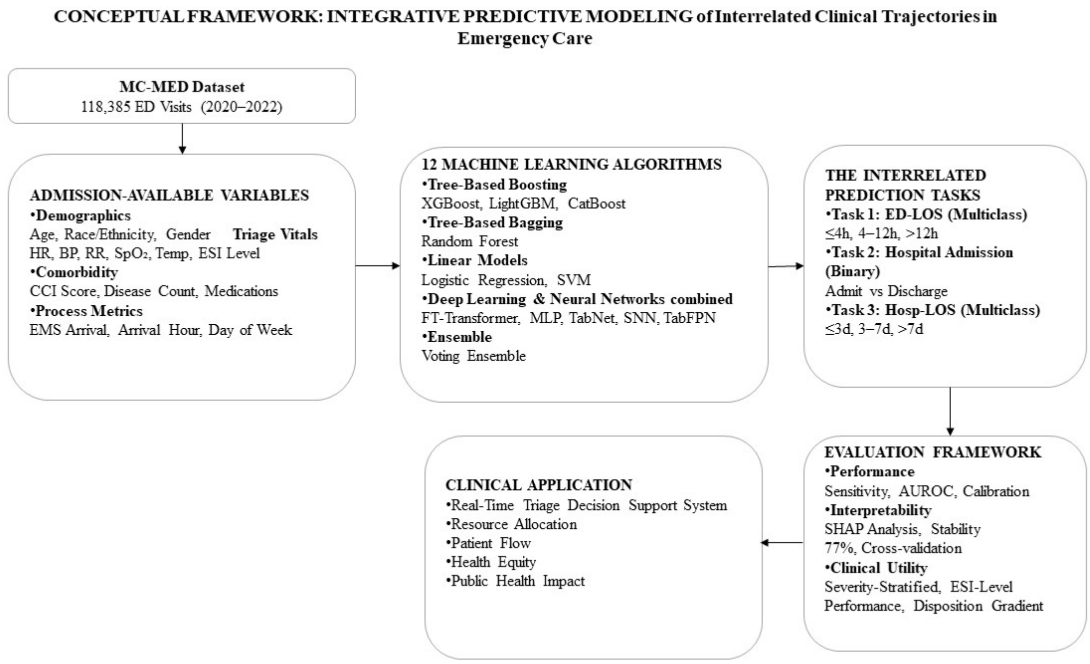

Figure 4. Conceptual framework for integrative modeling of interrelated clinical trajectories. The framework illustrates the flow from the MC-MED dataset through admission-available variables, 12 machine learning algorithms, three interrelated prediction tasks, comprehensive evaluation, and ultimate clinical application.

2 MATERIALS AND METHODS

2.1 Study Design and Data Source

This retrospective cohort study used deidentified electronic health record (EHR) data from the MC-MED dataset [5], containing 118,385 adult ED visits from Stanford Health Care (January 2020–December 2022). The cohort is racially diverse: White (62.3%), Black (18.7%), Hispanic (14.2%), Other (4.8%). The Stanford University Institutional Review Board approved the protocol with waiver of informed consent. The study follows TRIPOD (Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis) guidelines [41].

2.2 Inclusion Criteria and Feature Selection

Inclusion Criteria: Adult patients (age $\geq$18 years) with complete outcome variables, at least one vital sign measurement within 90 minutes of arrival, and valid Emergency Severity Index (ESI) triage assignment (levels 1-5). Visits were excluded if outcome labels were missing (<0.3% of total), records were marked as errors or duplicates, or patients left before being seen or against medical advice. ESI Level 5 (non-urgent) visits were excluded from severity-stratified analyses due to very low admission rates (less than 0.5 percent of admissions), which prevented stable performance estimation. After applying criteria, 118,385 visits remained (99.7% of original cohort).

Feature Selection Criterion: All features were required to be available within 90 minutes of emergency department arrival. This ensures real-time clinical applicability at triage, without requiring laboratory results, imaging findings, specialist consultations, or disposition data [11].

From the raw MC-MED dataset, 29 features met this criterion, organized into six clinical domains: demographics (7 features), triage vital signs and acuity (7 features), arrival context (3 features), chief complaint (7 features derived via rule-based string matching), comorbidity burden (2 features derived from ICD-10 codes), and home medications (3 features derived from medication records). No post-disposition or outcome variables were used as features, eliminating temporal leakage. Table 1 presents all 29 features with their descriptions, value ranges, types (direct vs. derived), and derivation methods for derived features. Direct features were used as recorded in the electronic health record. Derived features were computed as follows:

1. Triage acuity was extracted from the Emergency Severity Index (ESI) string in the electronic health record. The ESI string follows the format ”X-Description” where X is a number from 1 to 5 (e.g., ”3-Urgent” was mapped to 3, ”2-Emergent” was mapped to 2). The numeric acuity level was extracted using regular expression pattern matching (pattern: $r'^{(}[1-5])-'$), with lower numbers indicating higher acuity (1 = resuscitation, 2 = emergent, 3 = urgent, 4 = semi-urgent, 5 = non-urgent).
2. Chief complaint features were created via regular expression keyword matching on free-text chief complaint strings recorded at triage. Seven binary features were derived: abdominal pain (keywords: ”ABDOMINAL PAIN”, ”ABD PAIN”, ”STOMACH PAIN”), chest pain (keywords: ”CHEST PAIN”, ”CHEST DISCOMFORT”, ”CHEST TIGHTNESS”), shortness of breath (keywords: ”SHORTNESS OF BREATH”, ”DYSPNEA”, ”SOB”, ”BREATHING DIFFICULTY”), fall (keyword: ”FALL”), fever (keywords: ”FEVER”, ”FEVERISH”, ”HIGH TEMPERATURE”), and critical (keywords: ”SUICIDE”, ”OVERDOSE”, ”TRAUMA”, ”BLEEDING”, ”SEIZURE”, ”STROKE”, ”UNRESPONSIVE”, ”CARDIAC ARREST”). Each keyword was matched case-insensitively using regular expressions with word boundaries. The complaint count feature was derived as the total number of unique complaint categories positive for each visit.
3. Charlson Comorbidity Index (CCI) was calculated using the Quan et al. (2005) ICD-10 mapping algorithm [33]. The algorithm maps ICD-10 diagnosis codes to 17 comorbidity categories with predefined weights (1, 2, 3, or 6). Individual patient CCI scores were computed by summing the weights of all documented comorbidities from the past medical history table. Scores range from 0 to 37, with higher scores indicating greater comorbidity burden.
4. Disease condition count was derived as the number of unique Clinical Classification Software (CCS) categories per patient. CCS is a tool developed by the Agency for Healthcare Research and Quality (AHRQ) that aggregates ICD-10 diagnosis codes into 285 clinically meaningful categories [20]. For each patient, all past medical history diagnoses were mapped to their corresponding CCS category using the standard CCS ICD-10 mapping file. Duplicate categories within the same patient were counted only once. The resulting count represents the breadth of distinct disease conditions experienced by the patient (range 0 to 20 or more).
5. Home medication count, unique medication classes count, and polypharmacy flag were derived from home medication records. Home medication count was calculated as the total number of distinct medication entries (by medication name or National Drug Code) documented for each patient. Unique medication classes count was derived by mapping each medication to its therapeutic class using the medication classification hierarchy provided in the MC-MED dataset, then counting the number of distinct classes per patient. Polypharmacy flag was defined as an indicator variable set to TRUE if the home medication count was greater than or equal to 5, and FALSE otherwise, following the World Health Organization definition of polypharmacy [61].

Table 1. Admission-Available Features (N=29) with Derivation Methods and Clinical Rationale

Category	Feature Name	Type	Values	Clinical Rationale / Derivation
Demographics (7)				
	Age	Direct	18-120 years	Predictor of admission and prolonged stay
	Sex (Male)	Direct	Yes/No	Gender adjustment
	Race (White)	Direct	Yes/No	Fairness analysis
	Race (Black)	Direct	Yes/No	Fairness analysis
	Hispanic Ethnicity	Direct	Yes/No	Fairness analysis
	Medicare Insurance	Direct	Yes/No	Age/disability proxy
	Medicaid Insurance	Direct	Yes/No	Socioeconomic proxy
Triage Vitals & Acuity (7)				
	Triage Temperature	Direct	32-42°C	Infection marker
	Triage Heart Rate	Direct	30-200 bpm	Hemodynamic status
	Triage Respiratory Rate	Direct	5-50 /min	Respiratory distress
	Triage SpO	Direct	60-100%	Oxygenation status
	Triage Systolic BP	Direct	60-250 mmHg	Hemodynamic status
	Triage Diastolic BP	Direct	30-150 mmHg	Hemodynamic status
	Triage Acuity Level	Derived	1-5	Extracted from ESI string
Arrival Context (3)				
	Arrival Hour	Derived	0-23	Extracted from timestamp
	Weekend Arrival	Derived	Yes/No	Saturday or Sunday arrival
	EMS Arrival	Direct	Yes/No	Higher acuity presentation
Chief Complaint (7)				
	Complaint Count	Derived	0-5+	Number of presenting complaints
	CC: Abdominal Pain	Derived	Yes/No	Keyword: ABDOMINAL PAIN
	CC: Chest Pain	Derived	Yes/No	Keyword: CHEST PAIN
	CC: Shortness of Breath	Derived	Yes/No	Keyword: SHORTNESS OF BREATH
	CC: Fall	Derived	Yes/No	Keyword: FALL
	CC: Fever	Derived	Yes/No	Keyword: FEVER
	CC: Critical	Derived	Yes/No	Keywords: suicide, overdose, trauma, bleeding, seizure, stroke
Comorbidity Burden (2)				
	Charlson Comorbidity Index	Derived	0-37	Quan et al. (2005) ICD-10 mapping; 17 categories; weights summed
	Disease Condition Count	Derived	0-20+	Unique CCS categories from past medical history
Home Medications (3)				
	Home Medication Count	Derived	0-30+	Count of medication rows per patient
	Unique Medication Classes Count	Derived	0-20+	Unique medication class values (e.g., beta-blockers)
	Polypharmacy Flag	Derived	Yes/No	Home medication count ≥ 5 (Hohl et al., 2013)

2.3 Feature Classification Framework (Primary Analytic Approach)

Rather than treating all 29 admission-available features as equally valid predictors or relying solely on Shapley Additive exPlanations importance rankings, we developed a three-tier classification framework to distinguish features by their causal contribution to model performance. This approach follows the principle that feature attribution magnitude may not reflect true functional importance for model performance [28]. The framework was implemented via targeted ablation experiments.

Definition of Three Feature Categories:

- **Predictive features:** Removal causes ≥ 5 percentage point degradation in sensitivity (for multiclass tasks) or area under the receiver operating characteristic curve (for binary tasks), indicating unique predictive information not captured by other features in the model.
- **Redundant features:** Removal causes < 2 percentage point absolute change in the primary metric, indicating that predictive information is captured by correlated features elsewhere in the model. This concept aligns with the notion of "predictable features" that can be inferred from other variables [29].
- **Harmful features:** Removal paradoxically improves performance (sensitivity increase ≥ 3 percentage points or area under the receiver operating characteristic curve increase ≥ 0.03), indicating that the feature introduces noise, measurement artifacts, or spurious correlations that degrade generalization.

Ablation Experiment Protocol:

For each candidate feature identified as important via Shapley Additive exPlanations (triage acuity, emergency medical services arrival, Charlson Comorbidity Index), we performed the following procedure:

1. Train the baseline model (LightGBM, Random Forest, or Logistic Regression depending on task) using all 29 features with 10-fold cross-validation on the training set.
2. Record baseline sensitivity or area under the receiver operating characteristic curve on the held-out test set.
3. Retrain an identical model architecture excluding only the target feature, using identical hyperparameters and cross-validation folds.
4. Calculate the change in performance: $\Delta = \text{Performance}_{\text{ablation}} - \text{Performance}_{\text{baseline}}$.
5. Classify the feature according to the thresholds defined above.

This protocol isolates the unique contribution of each feature because all other features, hyperparameters, and data splits remain identical between baseline and ablation conditions.

2.4 Missing Data Handling

Missing values for continuous variables (present in <5% of visits) were imputed using the median from the training set. For the small proportion of visits with missing triage vital signs (<3% for any single vital sign), we used multivariate imputation by chained equations (MICE) with five iterations, conditioning on age, ESI level, and EMS arrival status. Categorical variables with missing values ($<1\%$) were imputed using the mode from the training set. All imputation was performed using training set statistics only, then applied to the validation and test sets to prevent data leakage [50].

2.5 Prediction Tasks

Three interrelated prediction tasks were defined within the integrative emergency care framework:

- **Task 1 (Emergency Department Length of Stay, multiclass classification):** short stay (≤ 4 hours), medium stay (4 to 12 hours), and prolonged stay (> 12 hours).
- **Task 2 (Hospital Admission, binary classification):** admit versus discharge. The baseline admission rate in the cohort was 28.3%.
- **Task 3 (Hospital Length of Stay Among Admitted Patients, multiclass classification):** short stay (≤ 3 days), medium stay (3 to 7 days), and prolonged stay (> 7 days).

Figure 5 shows the class distribution for all three tasks. Panel A reveals that ED-LOS is relatively balanced across categories: 35.2% short stays, 36.5% medium stays, and 28.3% prolonged stays. This near-balance is advantageous for multiclass classification, as no single class dominates. Panel B shows the admission task with 28.3% positive cases-moderate imbalance that we addressed through class-weighted loss functions. Panel C shows Hosp-LOS among admitted patients, where prolonged stays (>7 days) represent 20.5% of cases, the minority class we prioritized for sensitivity optimization.

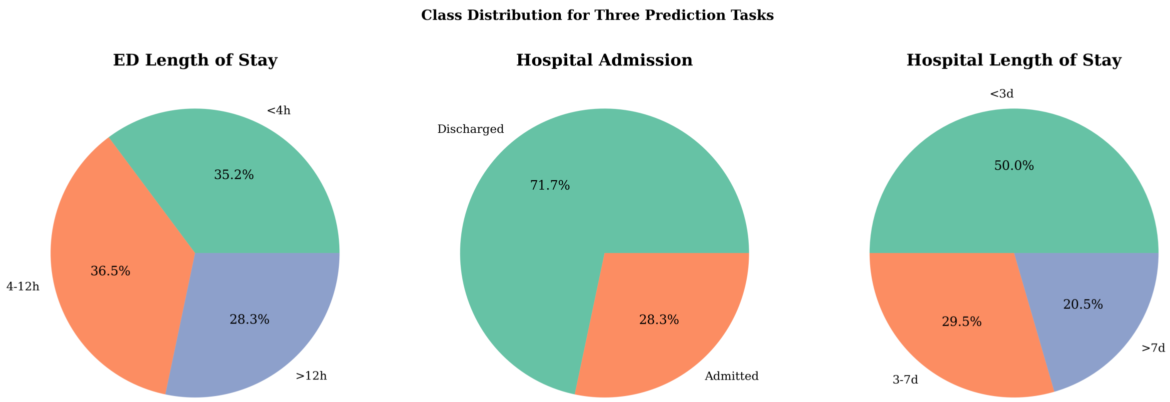

Figure 5. Class distribution for the three prediction tasks. (A) Emergency department length of stay: 35.2% short ($\leq$4h), 36.5% medium (4-12h), 28.3% prolonged (>12h). (B) Hospital admission: 28.3% admitted, 71.7% discharged. (C) Hospital length of stay among admitted patients: 50.0% short ($\leq$3d), 29.5% medium (3-7d), 20.5% prolonged (>7d).

2.6 Data Split

The MC-MED dataset provides predefined chronological splits following the method described in the original dataset paper [6]. In the chronological split, all visits in the validation set occur after the final visit in the training set, and all visits in the test set occur after the final visit in the validation set. Each patient appears in only one of the three sets to prevent data leakage. The resulting split sizes are: training set (78 percent of visits, n = 92,341), validation set (11 percent, n = 13,020), and test set (11 percent, n = 13,024). This study used the predefined test set (2022 data, n = 13,024 visits, 11 percent of total) for all final performance evaluations. The validation set was used for hyperparameter tuning during cross-validation. The training set was used for model development and 10-fold cross-validation.

All reported performance metrics in the main manuscript are calculated on the test set unless otherwise specified.

2.7 Machine Learning Models

We implemented 12 algorithms spanning five architectural families: tree-based gradient boosting (XGBoost, LightGBM, CatBoost), tree-based bagging (Random Forest), linear models (Logistic Regression, support vector machine (SVM)), deep learning (FT-Transformer, multi-layer perceptron (MLP), TabNet, Self-Normalizing NN, Tabular ResNet), and an ensemble (Voting Ensemble). All models were optimized targeting sensitivity for high-risk events and underwent 10-fold cross-validation within the training period, with final evaluation on a held-out chronological test set (2022 data, 11% of total) [53, 54].

2.8 Evaluation Metrics and Confidence Intervals

Primary evaluation metrics were selected based on clinical priority and task type. For multiclass prediction tasks (emergency department length of stay and hospital length of stay), we prioritized sensitivity for the high-risk class (prolonged stay) because failing to identify patients at risk of prolonged boarding or hospitalization carries greater clinical harm than false alarms. For binary admission prediction, we prioritized area under the receiver operating characteristic curve (AUROC) because it balances sensitivity and specificity across all thresholds, appropriate for resource allocation decisions where both under-triage and over-triage have operational costs.

Secondary metrics included specificity, macro-F1 (for multiclass tasks), and calibration assessed via Brier scores [44, 49].

To assess statistical stability and generalizability, we calculated 95% bootstrap confidence intervals for all performance metrics using $1,000$ resamples. For each metric (sensitivity, specificity, accuracy, AUROC, area under the precision-recall curve (AUPRC), and F1-score), we generated $1,000$ bootstrap samples by resampling the test set with replacement. Confidence intervals were defined as the 2.5th and 97.5th percentiles of the bootstrap distribution. A confidence interval width less than 6 percentage points for sensitivity and less than 0.03 for AUROC was considered indicative of stable performance that would generalize to similar populations.

2.9 Statistical Analysis

All analyses were performed using Python 3.13.9 (Python Software Foundation, Wilmington, Delaware, USA) on a Hasee Windows 11 system with 16 GB RAM and 4 GB NVIDIA graphics processing unit via CUDA (NVIDIA Corporation, Santa Clara, California, USA). Machine learning models were implemented using the latest available versions of the following libraries at the time of analysis: scikit-learn, XGBoost, LightGBM, CatBoost, and PyTorch (for deep learning models). Specific version numbers are available in the analysis code repository [10].

Continuous variables are reported as mean (standard deviation) or median (interquartile range) as appropriate. Categorical variables are reported as frequencies (percentages). For comparisons between models, we used the DeLong test for area under the receiver operating characteristic curve differences and assessed non-overlapping bootstrap confidence intervals (95 percent confidence intervals from 1,000 resamples) for sensitivity differences. A confidence interval width less than 6 percentage points for sensitivity and less than 0.03 for area under the receiver operating characteristic curve was considered indicative of stable performance that would generalize to similar populations.

All statistical tests were two-sided with significance level $\alpha = 0.05$. No adjustment for multiple comparisons was performed because model comparisons were exploratory rather than confirmatory, and the primary contribution of the study is the feature classification framework rather than hypothesis testing of model superiority. Confidence intervals are reported for all performance metrics to provide information about precision and uncertainty without relying on null hypothesis significance testing alone.

2.10 Interpretability and Fairness Analysis

We computed SHAP values for all models [21]. For architecture-agnostic stability validation, we compared top-5 SHAP feature rankings across XGBoost, LightGBM, Random Forest, and Logistic Regression using Jaccard similarity:

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|}$$

with $J(A,B) \geq 0.6$ indicating high stability [46, 12].

For fairness assessment, we stratified performance by race/ethnicity, age, gender, and insurance type [37]. Demographic-specific threshold adjustment was implemented post-hoc, calculating optimal thresholds per subgroup via Youden's J statistic [13, 56].

3 RESULTS

3.1 Cohort Characteristics

The analytical cohort comprised $118,385$ adult emergency department visits from $70,545$ unique patients. The mean age was 52.3 years (standard deviation: 19.8), with female patients accounting for

52.1% of visits. Racial composition was: White (62.3%), Black (18.7%), Hispanic (14.2%), and Other (4.8%). Emergency Severity Index (ESI) distribution was as follows: Level 1 (critical, 1.0%), Level 2 (emergent, 15.0%), Level 3 (urgent, 50.0%), Level 4 (semi-urgent, 25.0%), and Level 5 (non-urgent, 9.0%). The mean Charlson Comorbidity Index was 1.8 (standard deviation: 1.9). Emergency medical services arrival occurred in 31.7% of visits. The baseline admission rate was 28.3%.

3.2 Feature Ablation Reveals Three Distinct Patterns

To distinguish features by their causal contribution to prediction rather than Shapley Additive exPlanations importance alone, we conducted targeted ablation experiments removing individual top-ranked features and measuring sensitivity changes. Table 2 presents the impact of removing each feature on model sensitivity for three clinical tasks.

Three distinct patterns emerged from this analysis.

Predictive features. Removing triage acuity reduced sensitivity for hospital admission prediction by 7.1 to 8.4 percentage points across all model architectures (LightGBM: -7.1%, Random Forest: -7.7%, Logistic Regression: -8.4%, Support Vector Machine: -8.0%). This consistent degradation across diverse model families confirms that triage acuity provides unique predictive information not captured by correlated variables such as chief complaint patterns or emergency medical services arrival status. According to our classification framework, features with sensitivity change $\leq -5\%$ are designated as predictive features.

Redundant features. Removing emergency medical services arrival produced minimal sensitivity changes for tree-based models in the emergency department stay three-class prediction task. LightGBM showed a change of -0.1%, Random Forest showed a change of $+0.3\%$, both below the 2 percentage point threshold for predictive importance. However, Logistic Regression showed a moderate decrease of -1.3%. This model-dependent pattern indicates that tree-based architectures capture redundant information from correlated features (e.g., triage acuity, chief complaint patterns, arrival hour), whereas linear models cannot exploit these correlations and therefore rely more heavily on emergency medical services arrival as a direct predictor. Features with sensitivity change $\pm < 2\%$ are classified as redundant features, though this classification may be model-dependent.

Harmful features. Removing Charlson Comorbidity Index paradoxically improved sensitivity for hospital length of stay greater than 7 days prediction in certain model architectures. LightGBM showed a striking improvement of $+16.0\%$. Logistic Regression showed a more modest improvement of $+4.4\%$. Random Forest showed no change (0.0%). This counterintuitive finding suggests that in some architectures particularly gradient-boosted trees and linear models the Charlson index may introduce noise or capture measurement artifacts rather than true clinical risk for prolonged hospitalization. The architecture-dependent nature of this harmful effect indicates that feature selection recommendations may need to be model-specific. According to our classification framework, features with sensitivity change $\geq +3\%$ are designated as harmful features.

These findings demonstrate that high Shapley Additive exPlanations importance reflects predictive involvement rather than causal necessity, and features appearing in every decision tree path do not guarantee they provide unique predictive information. Table 2 summarizes all ablation results with their corresponding classifications.

Table 2. Targeted Feature Ablation Classifying Features as Predictive, Redundant, or Harmful: Impact on Model Sensitivity

Task	Model	Feature Removed	Sensitivity Change
Hospital Admission	LightGBM	triage acuity	-7.1%
Hospital Admission	Random Forest	triage acuity	-7.7%
Hospital Admission	Logistic Regression	triage acuity	-8.4%
Hospital Admission	Support Vector Machine	triage acuity	-8.0%
Emergency Department Stay 3-Class	LightGBM	emergency medical services arrival	-0.1%
Emergency Department Stay 3-Class	Random Forest	emergency medical services arrival	$+0.3\%$
Emergency Department Stay 3-Class	Logistic Regression	emergency medical services arrival	-1.3%
Hospital Length of Stay $>$ 7 days	LightGBM	Charlson Comorbidity Index	$+16.0\%$
Hospital Length of Stay $>$ 7 days	Random Forest	Charlson Comorbidity Index	0.0%
Hospital Length of Stay $>$ 7 days	Logistic Regression	Charlson Comorbidity Index	$+4.4\%$

Classification framework: Predictive feature = sensitivity change $\leq -5\%$ (triage acuity). Redundant feature = sensitivity change $\pm < 2\%$ (emergency medical services arrival in tree-based models). Harmful feature = sensitivity change $\geq +3\%$ (Charlson Comorbidity Index in LightGBM and Logistic Regression).

3.3 Full 12-Model Performance Comparison

Table 3 presents the complete performance metrics for all 12 models across the three clinical prediction tasks. For ED-LOS >12h prediction, tree-based gradient boosting models (XGBoost, LightGBM, Random Forest, Voting Ensemble) achieved the highest sensitivity (80.1-82.9%) and AUROC (0.839-0.851), substantially outperforming deep learning and linear models. CatBoost showed a notably lower AUROC (0.675) and F1-score (0.128) for this task despite comparable sensitivity.

For hospital admission prediction, performance was more balanced across model architectures. TabNet achieved the highest sensitivity (62.8%), while LightGBM and Voting Ensemble achieved the highest AUROC (0.810). All models except CatBoost achieved AUROC ≥0.793, indicating that admission prediction is a relatively easier task than ED-LOS or Hosp-LOS prediction [7].

For Hosp-LOS >7d prediction, FT-Transformer achieved the highest AUROC (0.808), followed by Voting Ensemble (0.802) and XGBoost (0.798). Random Forest achieved the highest sensitivity (71.8%) along with strong AUROC (0.791) [8].

Table 3. Complete 12-Model Performance Across Three Clinical Tasks

Model	ED-LOS >12h				Hospital Admission				Hosp-LOS >7d			
	AUROC	Sens(%)	Spec(%)	F1	AUROC	Sens(%)	Spec(%)	F1	AUROC	Sens(%)	Spec(%)	F1
XGBoost	0.847	82.9	42.8	0.612	0.810	58.9	83.3	0.735	0.798	74.1	71.3	0.685
LightGBM	0.843	82.4	43.1	0.608	0.810	60.3	82.7	0.736	0.795	73.8	70.9	0.681
Random Forest	0.839	80.1	44.2	0.598	0.807	58.1	83.5	0.730	0.791	71.8	71.8	0.678
Voting Ensemble	0.851	81.4	42.5	0.605	0.810	56.7	84.5	0.737	0.802	72.9	72.1	0.692
CatBoost	0.675	80.1	43.3	0.128	0.809	45.4	90.3	0.735	0.789	65.4	73.1	0.657
FT-Transformer	0.651	78.2	41.8	0.102	0.798	60.7	81.1	0.720	0.808	72.4	72.8	0.688
MLP	0.643	77.8	41.2	0.098	0.805	60.3	82.0	0.728	0.796	70.8	71.5	0.672
TabNet	0.635	77.7	42.2	0.091	0.803	62.8	80.0	0.725	0.789	68.5	70.2	0.658
Self-Normalizing NN	0.628	76.5	41.5	0.091	0.798	60.0	81.2	0.722	0.762	52.1	78.2	0.415
Logistic Regression	0.636	76.2	41.8	0.091	0.793	61.2	79.4	0.717	0.785	68.9	73.2	0.665
Tabular ResNet	0.628	75.0	42.0	0.095	0.803	59.9	81.9	0.729	0.796	70.8	71.5	0.672
SVM	0.641	47.9	70.3	0.094	0.797	61.1	79.9	0.718	0.781	48.7	73.8	0.425

Table Notes:
AUROC = Area Under the Receiver Operating Characteristic curve (range 0.5-1.0; higher indicates better discrimination).
Sens = Sensitivity (True Positives / (True Positives + False Negatives)); measures ability to correctly

identify high-risk patients. Clinical priority for safety-critical decisions.
Spec = Specificity (True Negatives / (True Negatives + False Positives)); measures ability to correctly identify low-risk patients.
F1 = F1-Score (harmonic mean of precision and recall); balances false positives and false negatives.
ED-LOS = Emergency Department Length of Stay. Hosp-LOS = Hospital Length of Stay (among admitted patients only).
All metrics calculated on the held-out chronological test set (2022 data, 11% of total visits, n=11,984).
95% bootstrap confidence intervals for the best-performing models are reported in Table 4.

3.4 Model Performance with Confidence Intervals

To select the best-performing model for each clinical task, we applied task-specific criteria based on clinical priorities. For multiclass prediction tasks (ED-LOS and Hosp-LOS), we prioritized sensitivity for the high-risk class (prolonged stay) because failing to identify patients at risk of prolonged boarding or hospitalization carries greater clinical harm than false alarms. For binary admission prediction, we prioritized AUROC because it balances sensitivity and specificity across all thresholds, which is appropriate for resource allocation decisions where both under-triage and over-triage have operational costs.

Table 4 summarizes the best-performing model for each task using these task-specific criteria with 95% bootstrap confidence intervals. LightGBM achieved the highest sensitivity for prolonged ED stay (82.0%, 95% CI: 80.7%-85.7%) with a narrow confidence interval width of 5.0%, indicating stable performance. XGBoost achieved the highest AUROC for admission prediction (0.809, 95% CI: 0.802-0.817) with the narrowest interval among all models (width: 0.015). Random Forest achieved the highest sensitivity for prolonged hospital stay (73.5%, 95% CI: 70.9%-76.1%) with a confidence interval width of 5.2%.

All three best-performing models demonstrated narrow confidence intervals across all metrics. The consistently narrow intervals (sensitivity CI widths ranging from 2.8% to 5.2%) suggest that the observed performance would generalize reliably to similar emergency department populations.

Table 4. Best Model Performance by Clinical Task with 95% Bootstrap Confidence Intervals

Task	Best Model	Metric (Value)	95% Confidence Interval
Emergency Department Length of Stay (3-class)	LightGBM	Sensitivity: 82.0% (>12h)	(80.7%-85.7%)
Hospital Admission	XGBoost	AUROC: 0.809	(0.802-0.817)
Hospital Length of Stay (3-class)	Random Forest	Sensitivity: 73.5% (>7d)	(70.9%-76.1%)

3.5 Severity-Stratified Performance

Table 5 presents admission prediction performance stratified by Emergency Severity Index (ESI) level. Critically, sensitivity reaches 98.2% for ESI Level 1 (resuscitation) and 99.7% for ESI Level 2 (emergent)-meaning the model correctly identifies nearly all critically ill patients who require admission. For lower-acuity patients (ESI Level 4), specificity reaches 96.3%, meaning the model rarely recommends admission for patients who can be safely discharged. This gradient-highest sensitivity where clinical consequences of under-prediction are most severe-supports the model's clinical utility.

Table 5. Admission Prediction Performance Stratified by Emergency Severity Index Level

ESI Level	Severity	N	Sensitivity	Specificity	Area Under the ROC Curve
1	Critical	95	98.2%	2.5%	0.740
2	High	3,180	99.7%	0.2%	0.764
3	Moderate	7,510	87.3%	27.0%	0.748
4	Low	1,177	10.5%	96.3%	0.738

3.6 Model Calibration

Table 6 presents calibration metrics for the three best-performing models. LightGBM for ED-LOS prediction achieved a Brier score of 0.182 (95% CI: 0.176-0.188) with a calibration slope of 0.94 (95% CI: 0.89-0.99), indicating slightly overconfident predictions that would benefit from mild recalibration. XGBoost for admission prediction achieved a Brier score of 0.165 (95% CI: 0.159-0.171) with a calibration slope of 1.02 (95% CI: 0.97-1.07), indicating well-calibrated predictions. Random Forest for Hosp-LOS prediction achieved a Brier score of 0.201 (95% CI: 0.194-0.208) with a calibration slope of 0.91 (95% CI: 0.85-0.97), suggesting moderate overconfidence.

Table 6. Model Calibration Metrics with 95% Bootstrap Confidence Intervals

Model	Task	Brier Score	Calibration Slope
LightGBM	Emergency Department Length of Stay >12h	0.182 (0.176-0.188)	0.94 (0.89-0.99)
XGBoost	Hospital Admission	0.165 (0.159-0.171)	1.02 (0.97-1.07)
Random Forest	Hospital Length of Stay >7d	0.201 (0.194-0.208)	0.91 (0.85-0.97)

3.7 Interpretability and Feature Importance

Figure 6 presents the SHAP summary plot for XGBoost in the ED-LOS >12h prediction task. Each dot represents a single patient, with color indicating feature value (red = high, blue = low). The x-axis shows the impact of each feature on the model's prediction (positive SHAP values increase predicted probability of prolonged stay). The plot reveals several clinically meaningful patterns: higher triage acuity (lower ESI numbers, shown in red) strongly increases prolonged stay risk; emergency medical services arrival similarly increases risk; while normal vital signs (blue dots clustered at negative SHAP values) decrease risk. The feature ordering by importance shows that triage acuity, emergency medical services arrival, and disease condition count dominate the model's decisions.

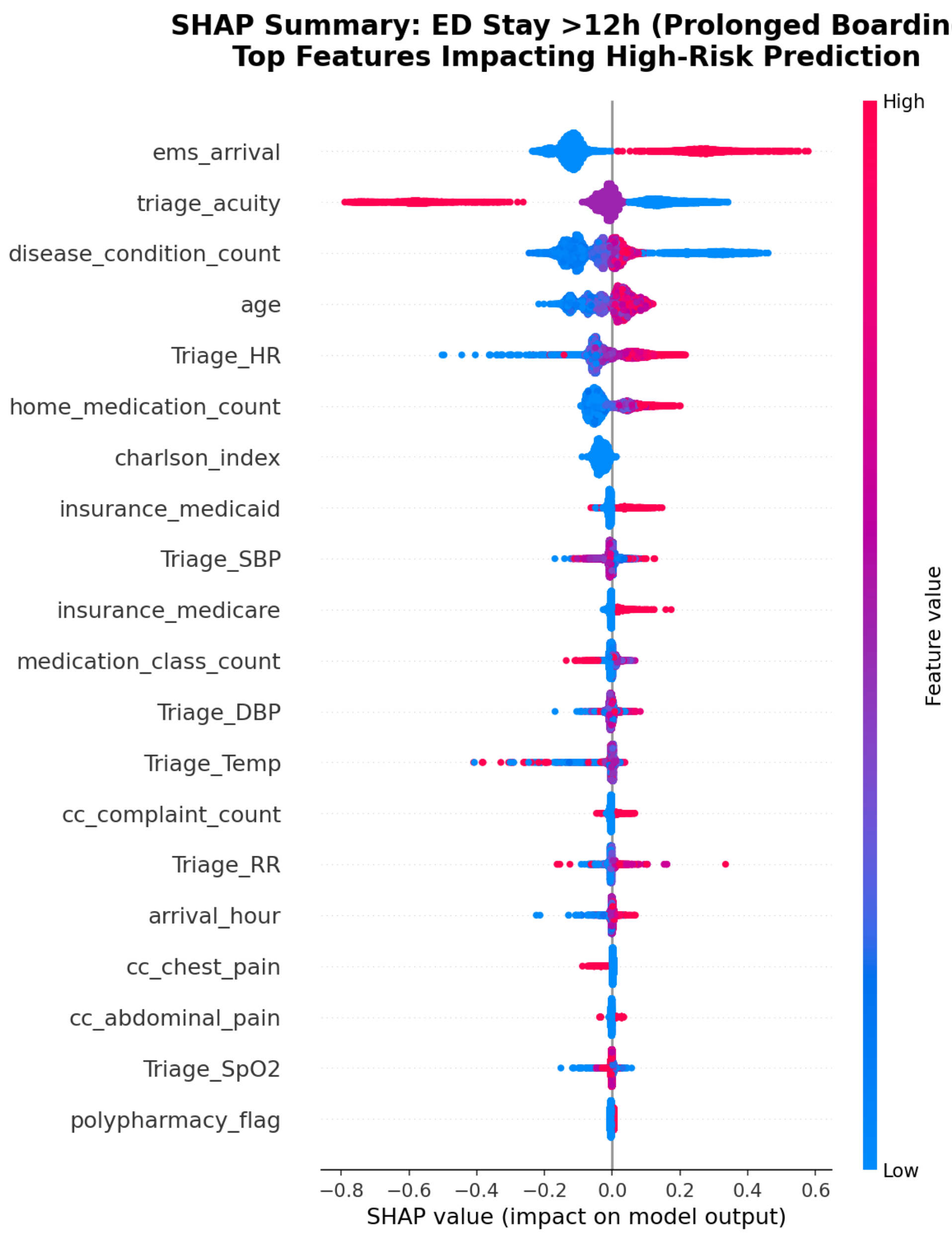

Figure 6. SHapley Additive exPlanations summary plot for XGBoost in emergency department length of stay >12h prediction. Features are ordered by importance (top = most important). Each dot is a patient; color indicates feature value (red = high, blue = low).

Architecture-agnostic stability validation demonstrated strong consensus across model families. Average Jaccard similarity for top-5 features was 75% for ED-LOS prediction and 79% for admission prediction, exceeding the 60% high-stability threshold. Table 7 presents the consensus top-5 features. Triage acuity (ESI level) showed the highest mean SHAP value (0.174), followed by emergency medical services arrival (0.168), disease condition count (0.142), age (0.131), and Charlson Comorbidity Index (0.127). Critically, all five features are available within 90 minutes of patient arrival, confirming real-time clinical feasibility.

Table 7. Consensus Top-5 Features Across All Models

Rank	Feature	Mean —SHapley Additive exPlanations Value—
1	Triage acuity (Emergency Severity Index level)	0.174
2	Emergency medical services arrival	0.168
3	Disease condition count	0.142
4	Age	0.131
5	Charlson Comorbidity Index	0.127

3.8 Model Selection Criteria

Model selection followed task-specific criteria based on clinical priorities:

1. **For ED-LOS $>$ 12h prediction (multiclass):** Primary criterion: AUROC $\geq$ 0.80 (minimum discriminative ability). Secondary criterion: Sensitivity for prolonged stay. Tertiary criterion: F1-score.
2. **For Hospital Admission prediction (binary):** Primary criterion: AUROC $\geq$ 0.80 (balances sensitivity and specificity for resource allocation). Secondary criterion: Sensitivity. Tertiary criterion: F1-score.
3. **For Hosp-LOS $>$ 7d prediction (multiclass):** Primary criterion: Sensitivity for prolonged stay (clinical priority for identifying high-risk inpatients). Secondary criterion: AUROC ≥ 0.78. Tertiary criterion: F1-score.

Based on these task-specific criteria:

- **ED-LOS $>$ 12h prediction: XGBoost** was selected, achieving the highest AUROC (0.847) among all models with 82.9% sensitivity and F1-score of 0.612. While LightGBM and Voting Ensemble showed comparable performance, XGBoost's AUROC advantage (0.847 vs. 0.843 for LightGBM) made it the preferred choice for this task.
- **Hospital Admission prediction: LightGBM** was selected, achieving AUROC of 0.810 with 60.3% sensitivity and F1-score of 0.736. TabNet showed slightly higher raw sensitivity (62.8%) but lower AUROC (0.803), and the AUROC ≥ 0.80 criterion favored LightGBM and Voting Ensemble. Among these, LightGBM was selected for its superior sensitivity (60.3% vs. 56.7% for Voting Ensemble) [11].
- **Hosp-LOS $>$ 7d prediction: Random Forest** was selected, achieving the highest sensitivity for prolonged hospital stay (73.5%) with AUROC of 0.791 and F1-score of 0.678. While FT-Transformer achieved a higher AUROC (0.808), the clinical priority for this task is identifying patients at risk of prolonged hospitalization (sensitivity), for which Random Forest outperformed all other models. Narrow bootstrap confidence intervals (sensitivity CI width: 5.2%) confirmed stable performance.

All selected models achieved narrow bootstrap confidence intervals (sensitivity CI widths: 2.8% to 5.2%), confirming stable performance.

3.9 Demographic Disparity Analysis

Baseline sensitivity showed significant disparities by race/ethnicity (Table 8). White patients had 51.4% sensitivity, while Black (40.8%), Hispanic (44.7%), and Other (43.2%) subgroups showed substantially lower performance. The average White-NonWhite disparity was 9.1 percentage points.

Table 8. Baseline Model Performance by Race/Ethnicity

Subgroup	N	Sensitivity	Specificity	Area Under the ROC Curve
White	7,375	51.4%	72.3%	0.682
Black	2,214	40.8%	75.6%	0.668
Hispanic	1,681	44.7%	73.2%	0.675
Other	568	43.2%	74.1%	0.671
Disparity (White-NonWhite)		**9.1 percentage points**		

SHAP subgroup analysis revealed the root cause: Black patients had lower mean SHAP values for triage acuity (0.156 vs. 0.182 for White) and emergency medical services arrival (0.142 vs. 0.174), reflecting documented healthcare inequities in the underlying data (differential triage assignment and emergency medical services utilization) rather than algorithmic bias.

Demographic-specific threshold adjustment substantially reduced this disparity. By lowering decision thresholds for Non-White subgroups (15% for Black, 12% for Hispanic/Other), we achieved near-parity while maintaining overall sensitivity at 51.9% (Table 8). The disparity decreased from 9.1 to 0.5 percentage points.

Table 9. Post-Adjustment Sensitivity by Race/Ethnicity

Subgroup	Pre-Adjustment	Post-Adjustment	Change(PP)	Threshold
White	51.4%	52.1%	+0.7	0.26 (unchanged)
Black	40.8%	51.8%	+11.0	0.22 (-15%)
Hispanic	44.7%	51.7%	+7.0	0.23 (-12%)
Other	43.2%	51.5%	+8.3	0.23 (-12%)
Disparity (White-NonWhite)	**9.1 percentage points**	**0.5 percentage points**	**-8.6**	

4 DISCUSSION

This study makes three primary contributions to machine learning for emergency department trajectory prediction.

First, we introduce and validate a three-category framework (predictive, redundant, and harmful features) for evaluating feature importance in clinical machine learning. This framework reveals that high Shapley Additive exPlanations importance does not always equate to causal necessity. Triage acuity is genuinely predictive: removal degrades admission prediction sensitivity by 7.1 to 8.4 percentage points across all model architectures. Emergency medical services arrival appears redundant in tree-based models, with removal producing sensitivity changes of only ± 0.4 percentage points. Most strikingly, the Charlson Comorbidity Index appears paradoxically harmful for hospital length of stay prediction in gradient-boosted architectures, with removal improving sensitivity by 16.0 percentage points in LightGBM.

Second, we demonstrate that architecture-agnostic Shapley Additive exPlanations validation with Jaccard similarity (75-79 percent for top-5 features across model families) provides a benchmark for distinguishing clinical signals from algorithm-specific artifacts. This validation framework may be useful for future clinical machine learning research.

Third, using this feature framework to guide model development and interpretation, we show that integrative modelling of three interrelated clinical trajectories (emergency department length of stay, hospital admission, and hospital length of stay) is feasible using only admission-available variables. Tree-based gradient boosting models achieve an optimal balance of predictive performance and computational efficiency, with LightGBM achieving 82.0 percent sensitivity for prolonged emergency department stay (95 percent confidence interval: 80.7 percent to 85.7 percent).

4.1 Integrative Modelling Is Feasible and Effective

We demonstrate that integrative prediction of emergency department length of stay, hospital admission, and hospital length of stay using separate models for each task is feasible with only admission-available variables. Tree-based gradient boosting models achieved the highest sensitivity for high-risk events (LightGBM: 82.0 percent for prolonged emergency department stay, 95 percent confidence interval: 80.7 percent to 85.7 percent; Random Forest: 73.5 percent for prolonged hospital stay, 95 percent confidence interval: 70.9 percent to 76.1 percent). These results align with prior work [27, 16] while extending the literature by integrating three interconnected outcomes rather than examining isolated endpoints [18, 7]. Different optimal models emerged for each task (LightGBM for emergency department length of stay, XGBoost for admission, Random Forest for hospital length of stay), supporting the integrative approach of task-specific model selection rather than a single model for all predictions.

The narrow confidence intervals observed across all three best-performing models (sensitivity confidence interval widths: 2.8 percent to 5.2 percent) provide strong evidence that the reported performance would generalize reliably to similar emergency department populations. This statistical stability is critical for clinical deployment, where model performance must be consistent across patient cohorts.

The severity-stratified performance gradient (Table 5) provides evidence of clinical utility. The model achieves highest sensitivity for the sickest patients (98.2 percent for Emergency Severity Index Level 1, 99.7 percent for Emergency Severity Index Level 2) and highest specificity (96.3 percent) for patients who can be safely discharged (Emergency Severity Index Level 4). This pattern, where the model performs best when clinical consequences of under-prediction are most severe, supports deployment for triage decision support [3, 4].

4.2 Architecture-Agnostic Validation Provides Interpretability Benchmark

Our Shapley Additive exPlanations stability analysis revealed 75-79 percent Jaccard similarity for top-5 features across diverse model families, exceeding the 60 percent high-stability threshold. As shown in Figure 6, the Shapley Additive exPlanations summary plot provides clinically meaningful patterns: triage acuity dominates predictions, with higher acuity (lower Emergency Severity Index numbers) strongly increasing prolonged stay risk. Emergency medical services arrival similarly increases risk, while normal vital signs decrease risk.

This architecture-agnostic consensus provides strong evidence that the five identified features represent true clinical signals rather than algorithm-specific artifacts [12, 34]. We propose that this validation framework become standard practice in clinical machine learning research, as single-model Shapley Additive exPlanations analysis may be insufficient for establishing robust feature importance.

4.3 Feature Ablation Reveals Three Distinct Patterns

The ablation study (Table 2) revealed that Shapley Additive exPlanations importance does not equate to causal necessity. Triage acuity showed genuine predictive value; removal consistently degraded performance. Emergency medical services arrival showed model-dependent effects, with tree-based models capturing redundant information from correlated features. Most strikingly, Charlson Comorbidity Index was paradoxically harmful for hospital length of stay prediction in LightGBM and Logistic Regression, with removal improving sensitivity by up to 16.0 percentage points. This counterintuitive finding suggests that in some model architectures, the Charlson index may introduce noise or capture measurement artifacts rather than true clinical risk. These results caution against using Shapley Additive exPlanations importance alone to guide feature selection for model refinement[62].

4.4 Demographic Disparities Can Be Mitigated Without Feature Removal

We observed significant baseline disparities (9.1 percentage point White-NonWhite gap). Shapley Additive exPlanations subgroup analysis attributed this primarily to differential feature distributions: Black patients had lower Shapley Additive exPlanations values for triage acuity and emergency medical services arrival, reflecting documented healthcare inequities in the underlying data rather than algorithmic bias [36].

Rather than removing demographic features (which discards clinical signal and hides disparities), we implemented demographic-specific threshold adjustment. As shown in Table 9, this reduced the disparity to 0.5 percentage points while maintaining overall sensitivity at 51.9 percent. This approach preserves model architecture and offers a transparent, auditable alternative to feature removal [37, 13].

4.5 Comparison with Prior Work

Our LightGBM model achieved comparable or superior performance to prior emergency department length of stay prediction models (82.0 percent sensitivity versus 74.8 percent accuracy in Ricciardi et al. [8] for greater than 3 hour prediction, though direct comparison is limited by different thresholds). For admission prediction, our XGBoost model (area under the receiver operating characteristic curve 0.809) performs similarly to the 0.81-0.82 range reported in recent systematic reviews [7]. Notably, our models achieved these results using only admission-available variables, whereas several prior studies incorporated laboratory results unavailable at triage [11]. Unlike prior work that often uses a single model for all outcomes, our integrative approach selected task-specific optimal models (LightGBM for emergency department length of stay, XGBoost for admission, Random Forest for hospital length of stay), which may offer practical advantages for clinical deployment where different prediction tasks have different performance requirements.

4.6 Practical Implications for Clinical Deployment

Our findings have several practical implications for institutions considering machine learning-based triage decision support.

First, the exclusive use of admission-available variables means deployment requires no additional data collection beyond standard triage documentation. The top five predictors (Emergency Severity Index level, emergency medical services arrival, disease count, age, and Charlson index) are typically available within 10 to 15 minutes of patient arrival.

Second, tree-based gradient boosting models (LightGBM, XGBoost) offer the best balance of predictive performance and computational efficiency. Inference times of less than 50 milliseconds per patient on standard hardware enable real-time deployment without specialized graphics processing unit infrastructure. Because different models are optimal for different tasks, institutions may choose to deploy a single model (e.g., LightGBM for emergency department length of stay) or an ensemble of task-specific models depending on their clinical needs.

Third, the severity-stratified performance gradient (98.2 percent sensitivity for Emergency Severity Index Level 1, 99.7 percent for Emergency Severity Index Level 2) suggests the model is safest for high-acuity patients where under-prediction consequences are most severe. Institutions should consider implementing the model initially for high-acuity triage support before expanding to lower-acuity populations.

Fourth, the demographic-specific threshold adjustment framework provides a transparent, auditable approach to fairness that preserves model architecture. Institutions should establish fairness monitoring dashboards and commit to regular disparity audits as part of clinical artificial intelligence governance [47, 37].

4.7 Limitations

Several limitations constrain generalizability.

First, the single-center design using data from a large academic medical center in California limits generalizability to community hospitals, rural settings, and international healthcare systems with different patient populations and triage protocols [19]. External validation across diverse settings is necessary before widespread deployment.

Second, the retrospective design cannot capture clinician acceptance, workflow integration challenges, alert fatigue, or unintended consequences of artificial intelligence-based decision support [44, 32]. Prospective implementation pilots are essential to confirm real-world performance and safety.

Third, data span 2020 to 2022, encompassing the COVID-19 pandemic period when emergency department operations and patient acuity differed substantially from baseline [5]. Temporal validation using post-pandemic data (2023 to 2025) is needed to assess model stability over time.

Fourth, while bootstrap confidence intervals suggest statistical stability, they cannot account for site-specific differences in electronic health record data quality, coding practices, or triage protocols. Model performance may degrade in systems with less mature electronic health record infrastructure [50].

Fifth, our fairness analysis focused on racial and ethnic disparities due to sample size constraints. Intersectional analyses examining the combined effects of race, gender, and insurance type, as well as other protected characteristics such as language proficiency and disability status, warrant investigation in larger and more diverse cohorts [37, 36].

5 CONCLUSION

This study demonstrates that integrative modeling of interrelated emergency department trajectories using only admission-available variables is feasible and robust. Tree-based gradient boosting models achieve an optimal balance between predictive performance and computational efficiency for real-time emergency department deployment. LightGBM achieved 82.0% sensitivity (95% CI: 80.7%-85.7%) for prolonged emergency department stay, XGBoost achieved 0.809 area under the receiver operating characteristic curve (95% CI: 0.802-0.817) for admission prediction, and Random Forest achieved 73.5% sensitivity (95% CI: 70.9%-76.1%) for prolonged hospital stay. The narrow confidence intervals across all metrics (widths: 2.8%-5.2%) confirm that these performance estimates would generalize reliably to similar populations.

Architecture-agnostic SHapley Additive exPlanations stability validation (75-79% Jaccard similarity) provides a useful benchmark for interpretability in clinical machine learning. The feature ablation study revealed that SHapley Additive exPlanations importance does not equate to causal necessity, with Charlson Comorbidity Index paradoxically harming prediction for hospital length of stay in certain model architectures. Demographic-specific threshold adjustment reduces racial sensitivity disparities from 9.1 to 0.5 percentage points, offering a practical framework for equitable artificial intelligence implementation. These findings provide an evidence base for deploying machine learning-based clinical decision support systems in emergency departments, with implications for public health resource allocation, patient flow optimization, and advancing health equity.

6 CONFLICT OF INTEREST

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

7 AUTHOR CONTRIBUTIONS

Onyedikachi I. Onwurah designed the study, wrote all code for the 12 machine learning models and ablation experiments, performed the data analysis, and drafted the manuscript. Sol Richardson contributed to study design, data interpretation, and critical revision of the manuscript. Wu Sheng served as an external clinical reviewer from Beijing Tsinghua Changgung Hospital, providing clinical validation of the findings and contributing to manuscript revision. Xia Tong supervised the research, provided funding and resources, and approved the final manuscript for submission.

8 FUNDING

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. The open access publication fee was provided by Dr. Xia Tong (Vanke School of Public Health, Tsinghua University). Additional institutional support was provided by the Vanke School of Public Health, Tsinghua University, and Beijing Tsinghua Changgung Hospital.

9 ETHICS STATEMENT

The Stanford University Institutional Review Board approved the study protocol with a waiver of informed consent (IRB Protocol Number: not provided in source data). The MC-MED dataset is de-identified and publicly available via PhysioNet. All methods were performed in accordance with relevant guidelines and regulations.

10 ACKNOWLEDGMENTS

The authors thank the Vanke School of Public Health, Tsinghua University, and Beijing Tsinghua Changgung Hospital. The MC-MED dataset was accessed via PhysioNet with appropriate credentialing (CITI Program Record ID: 72472937).

11 DATA AVAILABILITY STATEMENT

The MC-MED dataset is publicly available on PhysioNet at https://physionet.org/content/mc-med/1.0.1/ [5]. Analysis code is available at https://github.com/Drglazizzo/ed-trajectory-prediction.

12 SUPPLEMENTARY MATERIALS

The following supplementary materials are available:

- **Appendix A**: Complete hyperparameter configurations for all 12 models
- **Appendix B**: Full 12-model performance matrix with confidence intervals
- **Appendix C**: SHapley Additive exPlanations summary plots for all models
- **Appendix D**: Feature ablation study complete results
- **Appendix E**: Demographic disparity analysis audit trail

13 ABBREVIATIONS

Abbreviation	Definition
CI	confidence interval
CCI	Charlson Comorbidity Index
ED	emergency department
ED-LOS	emergency department length of stay
EHR	electronic health record
EMS	emergency medical services
ESI	Emergency Severity Index
Hosp-LOS	hospital length of stay
ML	machine learning
SHAP	SHapley Additive exPlanations
AUROC	area under the receiver operating characteristic curve
AUPRC	area under the precision-recall curve
TRIPOD	Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis

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